It is Taxing to be Coherent

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Introduction

- New Keynesian models subject to the ZLB generally do not posses a unique solution
- **Research question:** Can fiscal policy restore uniqueness?
- **Results:** Yes, even simple Ricardian fiscal policy can restore uniqueness.

This paper

- Uses the methodology in Gourieroux, Laffont, and Monfort (1980) (GLM) and Ascari and Mavroeidis (2022) (AM) to study minimum state variable (MSV) solution properties of NK models with fiscal policy
- Finds that active Ricardian fiscal policy can restore solution uniqueness
- Verifies that this result holds for fiscal policy that is financed via mix of taxes and debt
- Provides an algorithm for verifying existence and uniqueness of a solution for linearised models with an endogenous state

Example: Canonical NK model

- Take a baseline (log-linear) NK model as found in Galí (2015):

$$\text{DISE: } \hat{y}_t = E_t\hat{y}_{t+1} - \frac{1}{\sigma}(\hat{i}_t - E_t\hat{\pi}_{t+1}) + \varepsilon_t, \tag{1}$$

$$\text{NKPC: } \hat{\pi}_t = \beta E_t\hat{\pi}_{t+1} + \kappa\hat{y}_t + u_t, \tag{2}$$

$$\text{TR: } \hat{i}_t = \max\left\{-\mu, \phi_\pi\hat{\pi}_t + \phi_y\hat{y}_t\right\}, \tag{3}$$

where $\mu = \log(r\pi^*)$.

Illustrating the baseline NK model

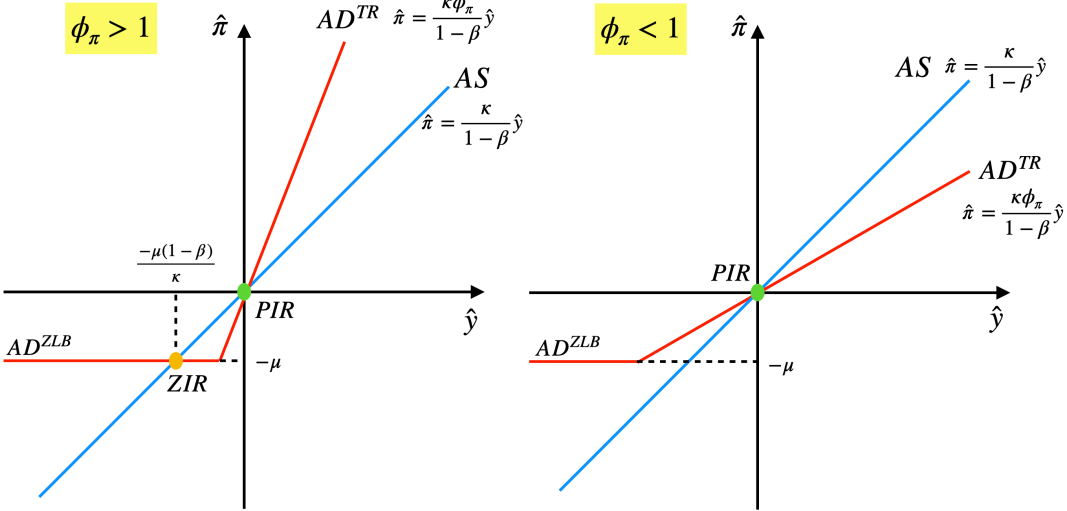


Figure The absorbing state for when $\varepsilon_t = 0$

Adding a fiscal authority

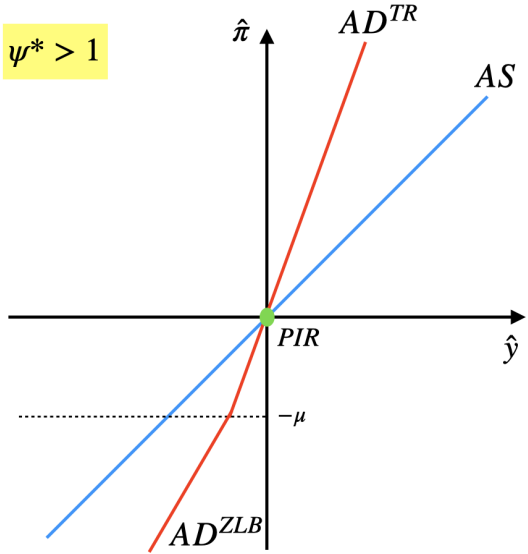
- Assume that government funds its spending with lump-sum and production taxes.

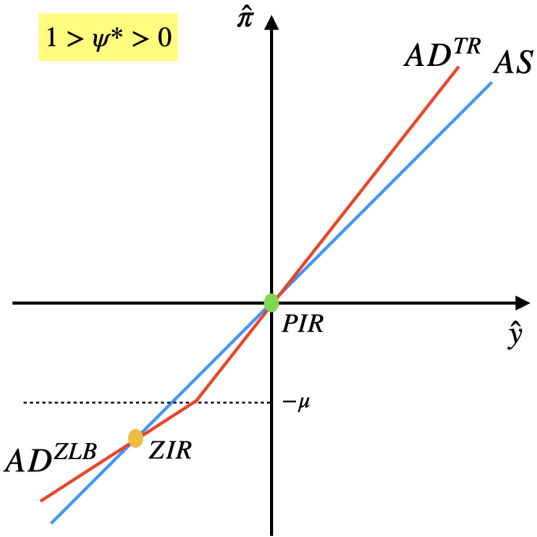
$$\tau_t = T_t + \tau_t^s Y_t = G_t \implies T\hat{t}_t + \tau^s Y(\hat{\tau}_t^s + \hat{y}_t) = G\hat{g}_t$$

- Assume $\hat{\tau}_t^s = \frac{\epsilon g}{c}\hat{g}_t$ and $\psi^* = \frac{\sigma g}{c}\psi_\pi$; model is given by

$$\begin{aligned}
\hat{y}_t &= E_t\hat{y}_{t+1} - \frac{c}{\sigma}\left(i_t - E_t\hat{\pi}_{t+1}\right) - \tau E_t \Delta\hat{\tau}_{t+1} + \varepsilon_t, \\
\hat{\pi}_t &= \beta E_t \hat{\pi}_{t+1} + \kappa_y \hat{y}_t + u_t, \\
\hat{i}_t &= \max\{-\mu, \phi_\pi \hat{\pi}_t + \phi_y \hat{y}_t\}, \\
E_t \Delta\hat{\tau}_{t+1} &= \psi_\pi \hat{\pi}_t + \psi_y \hat{y}_t.
\end{aligned} \tag{4}$$

CC with [simple] fiscal rules





Fiscal policy with government debt

- Government [nominal] budget constraint and [real] primary surplus are given by:

$$B_t = R_{t-1}B_{t-1} - P_t S_t,$$
$$S_t = T_t - G_t$$

- Define the real value of debt at maturity as:

$$B_t^m \equiv \frac{R_t B_t}{P_t}.$$

Rearrange to write the govt. nominal budget constraint as

$$\frac{B_t^m}{R_t} = \frac{B_{t-1}^m}{\Pi_t} - S_t. \tag{5}$$

- Fiscal policy block is closed by rules for T_t and G_t.

NK-FP model with government debt

$$
\begin{aligned}
\hat{x}_t &= E_t\hat{x}_{t+1} - \frac{c}{\sigma}(\hat{i}_t - E_t\hat{\pi}_{t+1} - \hat{r}_t^n) - gE_t\Delta\hat{g}_{t+1}, \\
\hat{\pi}_t &= \kappa_y\hat{x}_t - \kappa_g\hat{g}_t + \beta E_t\hat{\pi}_{t+1}, \\
\hat{b}_t^m &= \hat{i}_t + \frac{\hat{b}_{t-1}^m - \hat{\pi}_t - (1-\beta)\hat{s}_t}{\beta}, \\
\hat{t}_t &= \frac{\psi_b\hat{b}_{t-1}^m + \psi_x\hat{x}_t}{1-\beta}, \\
\hat{g}_t &= \frac{\psi_t^g\hat{b}_{t-1}^m + \psi_x^g\hat{x}_t + \psi_\pi^g\hat{\pi}_t}{1-\beta},
\end{aligned}
\tag{6}
$$

where now the natural interest rate is given by:

$$
\hat{r}_t^n = \frac{c}{\sigma}E_t\Delta\hat{y}_{t+1}^n.
$$

NK-FP model with government debt

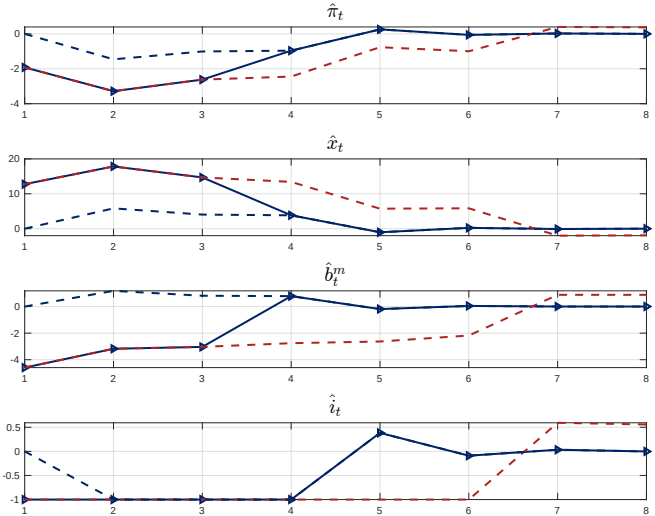


Figure Dynamic Solution (negative demand shock)

Calibration

Table Parameter Values and Descriptions

c	0.75	Fraction of consumption in output
g	0.25	Fraction of government spending in output
ϕ_π	1.5	Weight on inflation, Taylor Rule
ϕ_y	0.2	Weight on output gap, Taylor Rule
ψ_b	-0.01	FP parameter (transfers to debt)
ψ_x	-0.20	FP parameter (transfers to output)
ψ_b^g	-0.01	FP parameter (spending to debt)
ψ_x^g	-0.02	FP parameter (spending to output)
ψ_π^g	-0.10	FP parameter (spending to inflation)

CC conditions with endogenous state

- Along an MSV solution

$$E[\boldsymbol{Y}_{t+1}|\boldsymbol{Y}_t = Y\boldsymbol{e}_i] = E_t[\boldsymbol{Y}_{t+1}|\boldsymbol{X}_t = X\boldsymbol{e}_i] = Y\boldsymbol{K}^\top \boldsymbol{e}_i.$$

- Y must satisfy the following in order for the CC conditions to be satisfied:

$$\begin{aligned} 0 &= (\boldsymbol{A}_{s_i} Y + \boldsymbol{B}_{s_i} Y\boldsymbol{K}^\top + \boldsymbol{C}_{s_i} X + \boldsymbol{D}_{s_i} X\boldsymbol{K}^\top)\boldsymbol{e}_i, \\ s_i &= 1\{(\boldsymbol{a}^\top Y + \boldsymbol{b}^\top Y\boldsymbol{K}^\top + \boldsymbol{c}^\top X + \boldsymbol{d}^\top X\boldsymbol{K}^\top)\boldsymbol{e}_i > 0\},\ i = 1, \ldots, k. \end{aligned} \tag{7}$$

Testing the NK-FP model with debt for CC

$$\boldsymbol{A}_1 = \begin{pmatrix} -1 - \frac{\kappa_g}{1-\beta}\psi_\pi^g & \kappa_y - \frac{\kappa_g}{1-\beta}\psi_x^g & 0 \\ -\frac{c}{\sigma}\phi_\pi + \frac{g}{1-\beta}\psi_\pi^g & -1 + \frac{g}{1-\beta}\psi_x^g & -\psi_b^g\frac{g}{1-\beta} \\ -1 - \beta\phi_\pi - \psi_\pi^g & -\psi_x^g & \beta \end{pmatrix}, \boldsymbol{A}_0 = \begin{pmatrix} -1 - \frac{\kappa_g}{1-\beta}\psi_\pi^g & \kappa_y - \frac{\kappa_g}{1-\beta}\psi_x^g & 0 \\ \frac{g}{1-\beta}\psi_\pi^g & -1 + \frac{g}{1-\beta}\psi_x^g & -\psi_b^g\frac{g}{1-\beta} \\ -1 - \psi_\pi^g & -\psi_x^g & \beta \end{pmatrix},$$

$$\boldsymbol{B}_0 = \boldsymbol{B}_1 = \begin{pmatrix} \beta & 0 & 0 \\ \frac{c}{\sigma} - \psi_\pi^g\frac{g}{1-\beta} & 1 - \frac{g}{1-\beta}\psi_x^g & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \boldsymbol{D}^\top = \begin{pmatrix} -\frac{\kappa_g}{1-\beta}\psi_g^b & \frac{g}{1-\beta}\psi_b^g & -\frac{\kappa_g}{1-\beta}\psi_b^g \end{pmatrix},$$

$$\boldsymbol{C}_1 = \begin{pmatrix} 0 & 0 \\ -1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \boldsymbol{C}_0 = \begin{pmatrix} 0 & 0 \\ -1 & -\sigma\mu \\ 0 & 0 \end{pmatrix}.$$

Conclusion

- We verify that uniqueness and existence of an MSV solution for the NK-FP model with government debt.
- Resolves issue of satisfying Taylor principle and uniqueness of solution.
- Currently our algorithm is limited to models with one endogenous state.

References I

Ascari, Guido, and Sophocles Mavroeidis. 2022. "The Unbearable Lightness of Equilibria in a Low Interest Rate Environment." *Journal of Monetary Economics* 27:1–27.

Galí, Jordi. 2015. *Monetary Policy, Inflation, and the Business Cycle*. 2nd Edition. Princeton University Press.

Gourieroux, Christian, Jean-Jacques Laffont, and Alain Monfort. 1980. "Coherency Conditions in Simultaneous Linear Equation Models with Endogenous Switching Regimes." *Econometrica*, 675–695.

Appendix: Calibration

Table Parameter Values and Description

σ	2	Coefficient of relative risk-aversion
ζ	0.67	Frisch elasticity of labour supply
β	0.99	Discount factor
τ^c	0.2	Steady-state level of fiscal instrument
κ	0.23	NKPC coefficient
ψ^c	0.25	Coefficient on fiscal instrument
θ	0.75	Calvo probability of changing price
ε	10	Elasticity of substitution between goods
c	0.75	Fraction of consumption in output
g	0.25	Fraction of government spending in output
α_y	0.2	Output gap weight in Social Planner loss function
ϕ_π	{1.5,0.5}	Weight on inflation, Taylor Rule
ϕ_y	0.2	Weight on output gap, Taylor Rule